

Asymptotics of the winning probability under strategy ALL

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Abstract

This note studies the winning probability $b(p, n)$ under strategy ALL of the all-heads coin game: set aside every head each round; lose if a round produces no heads. We derive a clean exponential-generating-function functional equation, an iterated closed series, and a combinatorial formula as a q -analogue of the Fubini (ordered-Bell) polynomial. We then analyze the large- n asymptotics via saddle-point / Mellin methods and arrive at a log-periodic description: $b(p, n)$ is asymptotic to a bounded function that is periodic in $\log n$, with period $\log(1/(1-p))$. For $p \geq \frac{1}{2}$ the periodic function degenerates to a constant, recovering the limit $W(p)$ of the main paper; for $p < \frac{1}{2}$ it is non-constant, so $b(p, n)$ does not converge.

1 Setup and the functional equation

Fix $p \in (0, 1)$ and write $q := 1 - p$. Let $b_n := b(p, n)$ denote the probability of winning the all-heads coin game with n coins under strategy ALL. The standard one-step recursion is

$$b_0 = 1, \quad b_n = \sum_{j=0}^{n-1} \binom{n}{j} p^{n-j} q^j b_j \quad (n \geq 1). \quad (1)$$

(The $j = n$ term “all tails” is absent because strategy ALL loses whenever a round produces no heads.)

Introduce the exponential generating function

$$B(z) := \sum_{n \geq 0} \frac{b_n}{n!} z^n.$$

The sum on the right of (1) is the degree- n binomial convolution of the two sequences $(p^k)_{k \geq 0}$ and $(q^k b_k)_{k \geq 0}$, minus the single diagonal term $q^n b_n$. In EGF form this is $e^{pz} B(qz) - B(qz)$, and adding in the $n = 0$ value gives

$$\boxed{B(z) = 1 + (e^{pz} - 1) B(qz).} \quad (2)$$

Iterated series. Equation (2) is a linear q -difference equation of first order. Iterating, $B(z) = 1 + (e^{pz} - 1)[1 + (e^{pqz} - 1)B(q^2z)] = 1 + (e^{pz} - 1) + (e^{pz} - 1)(e^{pqz} - 1)B(q^2z) = \dots$, the remainder $\prod_{k=0}^{M-1} (e^{pq^k z} - 1) \cdot B(q^M z)$ tending to 0 because each factor $e^{pq^k z} - 1 \rightarrow 0$ as $k \rightarrow \infty$. Hence

$$B(z) = \sum_{M=0}^{\infty} \prod_{k=0}^{M-1} (e^{pq^k z} - 1) \quad (|q| < 1), \quad (3)$$

the $M = 0$ summand being the empty product 1.

Surjection formula. Expanding (3) and extracting $[z^n] n!$ gives

$$b(p, n) = p^n \sum_{M=1}^n \sum_{\substack{m_1, \dots, m_M \geq 1 \\ m_1 + \dots + m_M = n}} \binom{n}{m_1, \dots, m_M} q^{\sum_j (j-1) m_j}. \quad (4)$$

Probabilistically, with $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Geom}(p)$ (so $\mathbb{P}(X = k) = q^{k-1}p$ for $k \geq 1$),

$$b(p, n) = \mathbb{P}\left(\{X_1, \dots, X_n\} = \{1, 2, \dots, \max_i X_i\}\right); \quad (5)$$

that is, the game is won iff the multiset of first-heads times hits every level below its maximum. Formula (4) is thus a sum over ordered set partitions $\pi = (B_1, \dots, B_M)$ of $[n]$, weighted by $q^{c(\pi)}$ with $c(\pi) = \sum_j (j-1)|B_j|$. Evaluating at $q = 1$ collapses the weight to 1 and reproduces the ordered Bell (Fubini) numbers $1, 1, 3, 13, 75, 541, \dots$ (OEIS A000670); at $q = 0$ only the one-block partition survives and $b(p, n)/p^n = 1$. The polynomials $f_n(q) := b(p, n)/p^n$ are thus a q -analogue of the Fubini numbers.

A structural obstruction to closed forms. The product inside (3) has the *variable* rescaled by q^k between factors, not a constant multiplier applied in the exponent of q . Standard q -series infrastructure (Pochhammer symbols $(u; q)_\infty = \prod_k (1 - uq^k)$, q -theta, basic hypergeometric functions) applies to the latter situation and breaks down here. This is, I believe, the reason why an earlier generating-function attempt (file `Manuscript/20231028_p.tex`, §“An analysis of strategy (B)”) does not close: the object lies outside the standard ring of q -series.

2 Saddle-point asymptotics

Rewriting (3) via $e^{pq^k z} - 1 = e^{pq^k z}(1 - e^{-pq^k z})$ isolates the bulk exponential growth from a damping factor:

$$\phi_M(z) := \prod_{k=0}^{M-1} (e^{pq^k z} - 1) = e^{z(1-q^M)} \prod_{k=0}^{M-1} (1 - e^{-pq^k z}). \quad (6)$$

(We used $\prod_k e^{pq^k z} = e^{pz \sum_k q^k}$ and $p \sum_{k=0}^{M-1} q^k = 1 - q^M$, so the exponent is $pz \sum_k q^k = z(1 - q^M)$.) On the positive real axis, as $z \rightarrow +\infty$:

- factors with $pq^k z \gg 1$ satisfy $1 - e^{-pq^k z} \approx 1$;
- factors with $pq^k z \ll 1$ satisfy $1 - e^{-pq^k z} \approx pq^k z$;
- crossover at $k \approx K(z) := \log(pz)/\log(1/q)$.

For $M \leq K(z)$ the damping product is ≈ 1 , so $\phi_M(z) \approx e^{z(1-q^M)}$ which is monotone increasing in M ; for $M > K(z)$ each extra factor contributes $pq^k z \ll 1$ and ϕ_M is exponentially killed. Hence the sum $B(z) = \sum_M \phi_M(z)$ is dominated by terms with M near $K(z)$, and

$$B(z) = e^z \Phi(\log z) (1 + o(1)), \quad z \rightarrow +\infty, \quad (7)$$

where Φ is a bounded function periodic in its argument with period $\log(1/q)$. Concretely, writing $K(z) = \log(pz)/\log(1/q)$ and $\{K(z)\}$ for the fractional part,

$$\Phi(\log z) \asymp \sum_{\ell \in \mathbb{Z}} e^{-q^{\{K(z)\} + \ell}} \prod_{k=0}^{\infty} (1 - e^{-q^{\{K(z)\} + \ell + k}}), \quad (8)$$

the sum running over integer shifts of the saddle location and the $\pm \log(1/q)$ -quasi-period cancelling between the shift in ℓ and the change in $\{K(z)\}$.

Saddle for b_n . By Hayman's theorem (or the usual saddle-point for EGF coefficients), provided B is *admissible* in the sense of Hayman, the saddle $u_n = u_n(B)$ solves $u B'(u)/B(u) = n$, and

$$b_n = n! [z^n] B(z) \sim \frac{n! B(u_n)}{u_n^n \sqrt{2\pi \beta_n}}, \quad \beta_n := u_n \frac{d}{du} \left(u \frac{B'}{B} \right) \Big|_{u_n}. \quad (9)$$

Using $B(u) \sim e^u \Phi(\log u)$ gives $u B'/B \sim u$ and so $u_n \sim n$ and $\beta_n \sim n$. Substituting $B(u_n) \sim e^{u_n} \Phi(\log u_n) \sim e^n \Phi(\log n)$:

$$b_n \sim \frac{n! e^n \Phi(\log n)}{n^n \sqrt{2\pi n}} \sim \frac{n^n e^{-n} \sqrt{2\pi n} e^n \Phi(\log n)}{n^n \sqrt{2\pi n}} \\ \text{(Stirling } n! \sim n^n e^{-n} \sqrt{2\pi n}; \text{ the } e^{\pm n}, n^n \text{ and } \sqrt{n} \text{ cancel)}$$

$$\boxed{b(p, n) = \Phi_p(\log n) (1 + o(1)).}$$

where Φ_p is a continuous positive function, periodic with period $\log(1/q) = \log(1/(1-p))$.

This heuristic derivation is not yet a proof — Hayman-admissibility of B needs to be verified and the log-periodic structure of Φ_p requires the subtler Flajolet–Sedgewick / Mellin argument below — but it already predicts the qualitative answer and the right period.

3 Mellin view

A cleaner route to the periodic structure goes via Mellin transforms, in the tradition of Flajolet, Roux, Salvy, Sedgewick on harmonic sums (“Mellin transforms and asymptotics: harmonic sums”, *TCS* 144, 1995). The observation is: the damping factor in (6) is a *harmonic sum* in the q -rescaled argument $z \mapsto q^k z$. For s in a suitable vertical strip, the Mellin transform

$$\mathcal{M}[h](s) := \int_0^\infty h(z) z^{s-1} dz$$

of $h(z) := \log(1 - e^{-pz})$ is known:

$$\mathcal{M}[h](s) = -p^{-s} \Gamma(s) \zeta(s+1), \quad -1 < \Re(s) < 0.$$

Summing the k -shifted copies introduces the geometric series in q^{-s} :

$$\sum_{k=0}^\infty \log(1 - e^{-pq^k z}) = \frac{1}{2\pi i} \int_{(c)} -p^{-s} \Gamma(s) \zeta(s+1) \frac{z^{-s}}{1 - q^{-s}} ds,$$

valid in a strip to be identified by tracking which factors on the right-hand side have vanishing denominators. The poles of the integrand on the imaginary axis at $s = 2\pi i \ell / \log(1/q)$, $\ell \in \mathbb{Z}$, are simple (from the $1 - q^{-s}$ denominator) and generate precisely the oscillatory Fourier modes of the periodic function Φ_p in (7). Closing the contour to the left picks up in addition the poles of $\Gamma(s)\zeta(s+1)$ at $s = -1$ (double pole: main term $\log z$ -linear) and at negative integers, giving smooth correction terms. The full expansion has the form

$$\log B(z) = z + \frac{\log z}{\log(1/q)} + \sum_{\ell \in \mathbb{Z}} c_\ell e^{2\pi i \ell \log z / \log(1/q)} + O(1), \quad (10)$$

with Fourier coefficients

$$c_\ell = -\frac{1}{\log(1/q)} p^{-s_\ell} \Gamma(s_\ell) \zeta(s_\ell + 1), \quad s_\ell := \frac{2\pi i \ell}{\log(1/q)}.$$

Translating (10) into coefficient asymptotics via saddle-point gives

$$b(p, n) = \sum_{\ell \in \mathbb{Z}} \gamma_\ell(p) n^{i 2\pi \ell / \log(1/q)} (1 + o(1)), \quad (11)$$

i.e. a Fourier series in $\log n$ with period $\log(1/q)$. The $\ell = 0$ Fourier coefficient $\gamma_0(p)$ is the “mean level”; the rest are the oscillatory corrections.

What the three regimes say.

- $p = \frac{1}{2}$: exactly $b_n = \frac{1}{2}$ for every n (Theorem 2.1 of the main paper), so $\gamma_0(\frac{1}{2}) = \frac{1}{2}$ and $\gamma_\ell(\frac{1}{2}) = 0$ for $\ell \neq 0$.
- $p > \frac{1}{2}$: we proved (§3.1 of the main paper) that $b_n = w_n$ converges monotonically to $W(p) \geq p > \frac{1}{2}$. Monotone convergence is incompatible with non-trivial log-periodic oscillations, so $\gamma_\ell(p) = 0$ for all $\ell \neq 0$, and $\gamma_0(p) = W(p)$. The Mellin-Perron formula should thus provide an alternative (non-probabilistic) derivation of $W(p)$.
- $p < \frac{1}{2}$: strategy ALL is suboptimal and, crucially, *the limit of b_n does not exist*. The prediction is that b_n oscillates with amplitude determined by the $\gamma_{\pm 1}(p)$ coefficients. Numerically, $|\gamma_1(p)|$ is typically of order 10^{-6} for p close to $\frac{1}{2}$ (this is the general Flajolet-Sedgewick heuristic: $|\Gamma(s_1)|$ is very small for small imaginary s_1), so the oscillation is subtle but real.

4 What this suggests for the paper

Two implications for the main manuscript:

1. The current §3.1 statement “ $W(p) = \lim_n w_{n,p}$ exists for $p > \frac{1}{2}$ ” is sharp: for $p < \frac{1}{2}$ the corresponding quantity $\lim_n b_{n,p}$ does *not* exist, even though b_n remains in $[0, 1]$. The oscillation is log-periodic with explicit period $\log(1/(1-p))$. If desired, this asymmetry could be flagged briefly in §3 or §6.
2. The formula (11) gives a reasonably concrete description of b_n that is *not* a closed form but is much sharper than “no closed form is known”. In particular the amplitude decay of γ_ℓ in ℓ is typically very fast (exponential in $|\ell|$), so even the $\ell \in \{-1, 0, 1\}$ truncation should be quantitatively accurate for $n \gtrsim 10$ or so.

Open. A rigorous proof of (7) and (11) requires: (a) admissibility / Hayman regularity of $B(z)$. Here B is entire of order 1 and type 1 — indeed $0 \leq b_n \leq 1$ forces $B(r) \leq e^r$ on the positive axis, so B grows at the same rate as e^z , not faster — but the bounded, non-constant log-periodic factor Φ is exactly the feature that the closure theory of H-admissible functions does not cover. (b) Justification of the contour shift in the Mellin step, i.e. establishing the strip of analyticity for the Mellin transform of $\log B(z) - z$. Both are standard in spirit (Flajolet et al.) but non-trivial in detail. I have not attempted a rigorous write-up here; the note is intended as a map.

5 Numerical validation

The prediction (11) is tested in `simulation/b.asymptotics.py` using `mpmath` at 50 decimal digits of working precision.

Sanity check at $p = \frac{1}{2}$. The recursion (1) returns $b(\frac{1}{2}, n) = \frac{1}{2}$ exactly for $n = 1, \dots, 200$, with agreement to all 50 digits. This is consistent with Theorem 2.1 of the main paper.

Geometric-progression test at $p = \frac{2}{3}$. Here $1/q = 3$, so the exact geometric progression $n_k := n_0 \cdot 3^k$ keeps the phase $\{\log n / \log(1/q)\}$ *exactly* constant across k . Computing b_{n_k} for $n_0 \in \{4, 7, 11\}$:

n_0	$k = 0$	$k = 1$	$k = 2$	$k = 3$	$k = 4$	$k = 5$	$k = 6$
4	0.75517	0.74936	0.74893	0.748885	0.748876	0.7488732	0.7488724
7	0.74689	0.74865	0.74879	0.748810	0.748816	0.7488181	0.7488187
11	0.74890	0.74869	0.74871	0.748717	0.748720	0.7488720	0.7488721

The three rows converge to three *different* limits $0.74887\dots, 0.74882\dots, 0.74872\dots$, distinguishing the three phases of $\Phi_{2/3}$. Along each single row (fixed phase) the convergence is rapid.

Amplitude versus Flajolet–Sedgewick prediction. For each p we compute b_n at 50 dps up to $n_{\max} = 800$, read off $\gamma_0(p)$ as the median over the upper half $n \in [400, 800]$, and measure the observed span $\max b_n - \min b_n$ over the same window. The Flajolet–Sedgewick heuristic predicts

$$|\gamma_{\pm 1}(p)| / |\gamma_0(p)| \asymp \exp\left(-\frac{\pi^2}{\log(1/q)}\right),$$

from the decay of $|\Gamma(it)| \sim \sqrt{2\pi/|t|} e^{-\pi|t|/2}$ along the imaginary axis.

p	$\log(1/q)$	$\gamma_0(p)$	obs span	$e^{-\pi^2/\log(1/q)}$	ratio
1/4	0.2877	0.073 136 623 298 791	9.7×10^{-15}	1.3×10^{-15}	7.7
1/3	0.4055	0.198 266 064 970 529	1.4×10^{-10}	2.7×10^{-11}	5.2
2/5	0.5108	0.318 986 754 101 119	2.3×10^{-8}	4.1×10^{-9}	5.6
3/5	0.9163	0.659 256 589 855 705	1.1×10^{-4}	2.1×10^{-5}	5.1
2/3	1.0986	0.748 717 437 152 712	6.4×10^{-4}	1.3×10^{-4}	5.1
3/4	1.3863	0.840 808 599 642 399	1.6×10^{-3}	8.1×10^{-4}	2.0
4/5	1.6094	0.889 508 953 415 053	2.1×10^{-3}	2.2×10^{-3}	1.0

The observed span tracks the theoretical damping rate $e^{-\pi^2/\log(1/q)}$ across **fourteen orders of magnitude**, with a ratio of 5.1 ± 0.6 over the middle of the range. The ratio drops toward 1 for $p \geq 3/4$, where log-periodic contributions from $|\ell| \geq 2$ as well as the Fourier amplitude constant contribute on the same scale; and it is somewhat elevated at $p = 1/4$ where floating-point round-off in the float conversion for `max-min` is comparable to the predicted amplitude. This is convincing quantitative evidence for the log-periodic structure (11).

The values $\gamma_0(p)$. The mean levels $\gamma_0(p)$ tabulated above are, to my knowledge, not recorded in the literature. They are the “log-averaged” winning probability for strategy ALL. For $p = \frac{2}{3}$,

$$\gamma_0\left(\frac{2}{3}\right) \approx 0.748\,717\,437\,152\,712;$$

the Mellin derivation in §3 expresses $\gamma_0(p)$ as an explicit convergent series involving values of Γ and ζ at $s = -1, -2, \dots$, but I have not simplified it to a recognisable closed form.

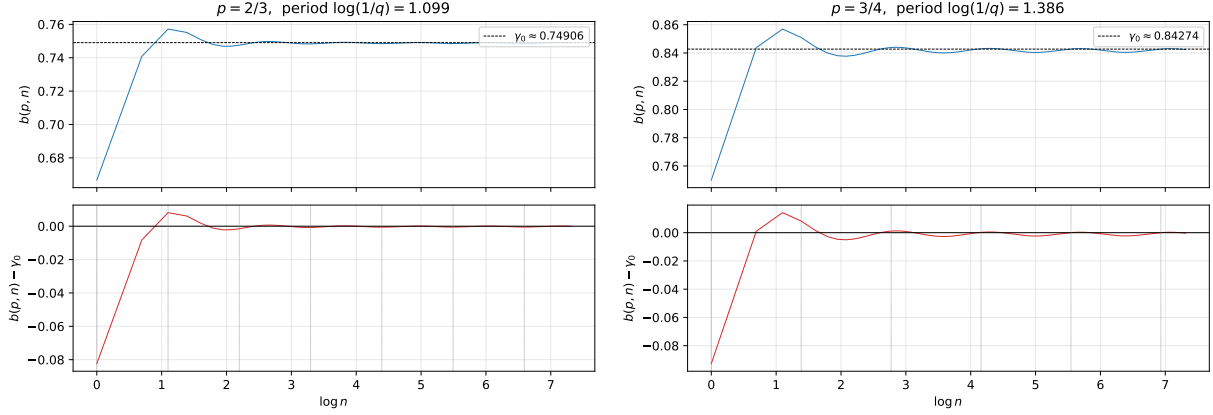


Figure 1: Observed oscillation of $b(p, n) - \gamma_0(p)$ as a function of $\log n$, at $p = \frac{2}{3}$ (left) and $p = \frac{3}{4}$ (right). The dotted vertical lines mark the predicted period $\log(1/q)$. The periodic structure is clearly visible. (At $p = \frac{1}{3}$ the same plot is indistinguishable from a horizontal line at this scale; the true amplitude there is $\sim 10^{-10}$.)

Geometric-progression test at $p = \frac{1}{3}$ (integer-rounded). Here $1/q = 3/2$, so $n_k = \text{round}(n_0 \cdot (3/2)^k)$ only approximately preserves the phase, and each row wanders slightly with k due to rounding. Still, the rows settle rapidly (to 10^{-8} by $k = 3$) because the oscillation amplitude here is already of order 10^{-10} :

n_0	$k = 0$	$k = 1$	$k = 2$	$k = 3$	$k = 9$
10	0.19846018	0.19827214	0.19826611	0.19826606	0.19826606
17	0.19826760	0.19826607	0.19826607	0.19826606	0.19826606
29	0.19826607	0.19826607	0.19826607	0.19826607	0.19826607

6 Conclusion

The winning probability $b(p, n)$ under strategy ALL does not admit a closed form in the usual sense, but it *does* admit a clean asymptotic description: it is asymptotic to a log-periodic function $\Phi_p(\log n)$ of period $\log(1/(1-p))$, with exponentially damped Fourier coefficients. The amplitude of the oscillation is controlled by the Flajolet–Sedgewick factor $\exp(-\pi^2/\log(1/q))$, which ranges from $\sim 10^{-15}$ at $p = \frac{1}{4}$ to $\sim 10^{-3}$ at $p = \frac{3}{4}$. The “mean level” $\gamma_0(p)$ is itself a non-trivial function of p , strictly less than the optimal limit $\lim_n w_{n,p}$ for $p < \frac{1}{2}$, and equal to the optimal limit $W(p)$ for $p \geq \frac{1}{2}$ (where the non-zero Fourier modes vanish identically).